

BLDC_Controller - Technical User Manual

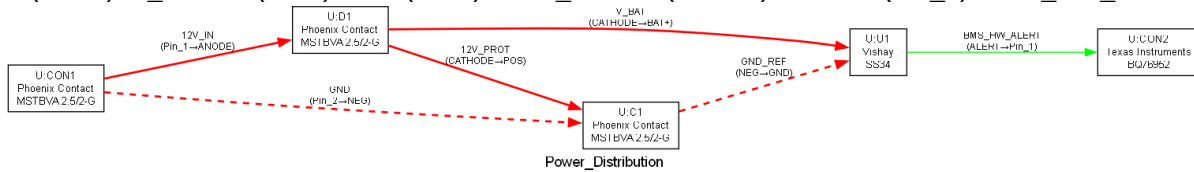
1. Bill of Materials

IDs	Make	Model	Part_Number	Type	Description
	Infineon	IMC300A	IMC300A	Power Management IC	Power Management IC< im_end >
	Murata	GRM31CR71E106KA12	GRM31CR71E106KA12	Capacitor	Multilayer Ceramic Capacitor< im_end >
	Phoenix Contact	MSTBVA 2.5/2-G	MSTBVA 2.5/2-G	Terminal Block	Terminal block, 2.5 mm² cross-section, 2-pole, with spring co
	STMicroelectronics	STM32G474	STM32G474	Microcontroller	32-bit ARM Cortex-M4F Microcontroller< im_end >
	Texas Instruments	BQ76952	BQ76952	IC	Power Management IC, Battery Management
	Vishay	SS34	SS34	Resistor	0.25W, 1% Metal Film Resistor< im_end >
	Yageo	RC0402FR-0710KL	RC0402FR-0710KL	Resistor	0710kOhm, 0402 package

2. Subsystem Descriptions & Diagrams

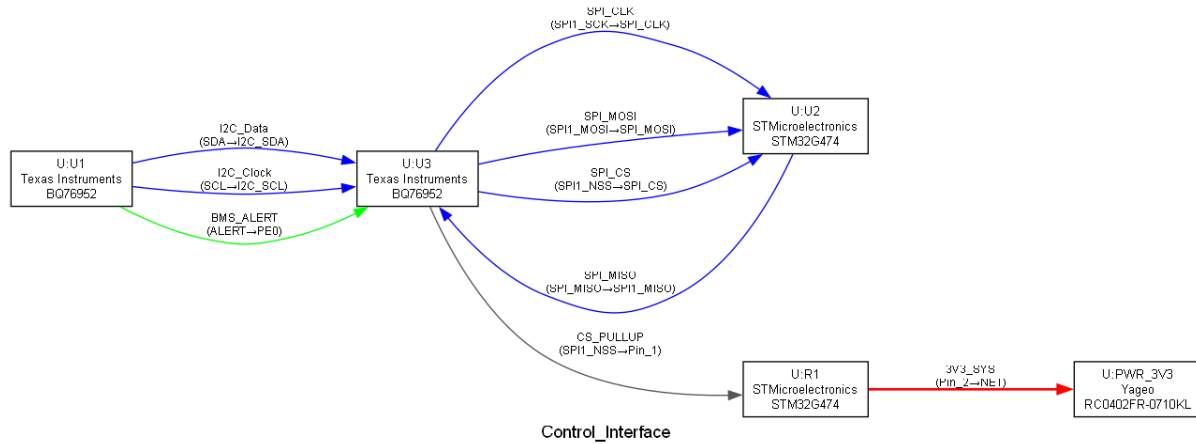
Power_Distribution

****Power_Distribution - Functional Description**** ****Overview:**** The Power_Distribution subsystem consists of 5 components connected via 6 unique signals. ****Components:**** - CON1: Phoenix Contact MSTBVA 2.5/2-G - D1: Phoenix Contact MSTBVA 2.5/2-G - C1: Phoenix Contact MSTBVA 2.5/2-G - U1: Vishay SS34 - CON2: Texas Instruments BQ76952 ****Signals:**** - 12V_IN - GND - 12V_PROT - V_BAT - GND_REF - BMS_HW_ALERT ****Signal Flow:**** CON1(Pin_1) -> D1(ANODE): 12V_IN CON1(Pin_2) -> C1(NEG): GND D1(CATHODE) -> C1(POS): 12V_PROT D1(CATHODE) -> U1(BAT+): V_BAT C1(NEG) -> U1(GND): GND_REF U1(ALERT) -> CON2(Pin_1): BMS_HW_ALERT



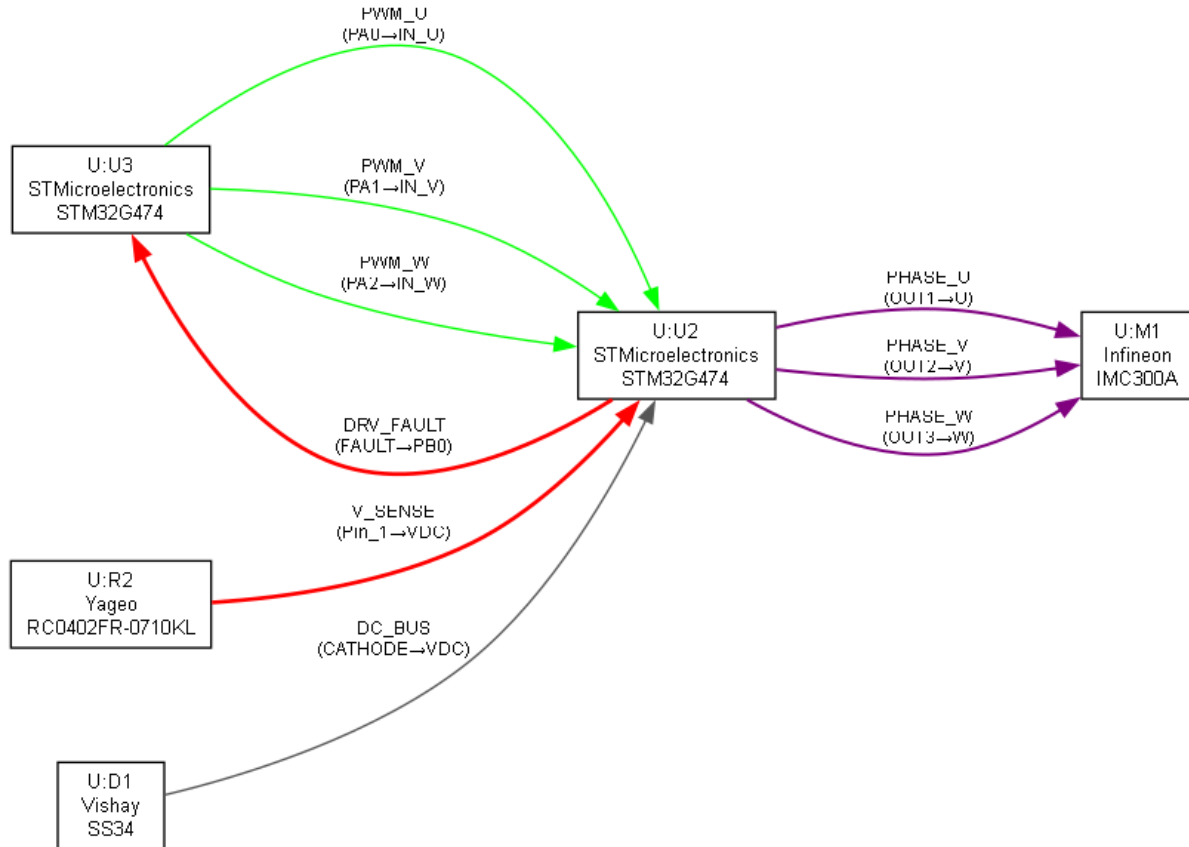
Control_Interface

****Control_Interface - Functional Description**** ****Overview:**** The Control_Interface subsystem consists of 5 components connected via 9 unique signals. ****Components:**** - U1: Texas Instruments BQ76952 - U3: Texas Instruments BQ76952 - U2: STMicroelectronics STM32G474 - R1: STMicroelectronics STM32G474 - PWR_3V3: Yageo RC0402FR-0710KL ****Signals:**** - I2C_Data - I2C_Clock - BMS_ALERT - SPI_CLK - SPI_MOSI - SPI_MISO - SPI_CS - CS_PULLUP - 3V3_SYS ****Signal Flow:**** U1(SDA) -> U3(I2C_SDA): I2C_Data U1(SCL) -> U3(I2C_SCL): I2C_Clock U1(ALERT) -> U3(PE0): BMS_ALERT U3(SPI1_SCK) -> U2(SPI_CLK): SPI_CLK U3(SPI1_MOSI) -> U2(SPI_MOSI): SPI_MOSI U2(SPI_MISO) -> U3(SPI1_MISO): SPI_MISO U3(SPI1_NSS) -> U2(SPI_CS): SPI_CS U3(SPI1_NSS) -> R1(Pin_1): CS_PULLUP R1(Pin_2) -> PWR_3V3(NET): 3V3_SYS



Motor_Drive

****Motor_Drive - Functional Description**** ****Overview:**** The Motor_Drive subsystem consists of 5 components connected via 9 unique signals. ****Components:**** - U3: STMicroelectronics STM32G474 - U2: STMicroelectronics STM32G474 - M1: Infineon IMC300A - R2: Yageo RC0402FR-0710KL - D1: Vishay SS34 ****Signals:**** - PWM_U - PWM_V - PWM_W - DRV_FAULT - PHASE_U - PHASE_V - PHASE_W - V_SENSE - DC_BUS ****Signal Flow:**** U3(PA0) -> U2(IN_U): PWM_U U3(PA1) -> U2(IN_V): PWM_V U3(PA2) -> U2(IN_W): PWM_W U2(FAULT) -> U3(PB0): DRV_FAULT U2(OUT1) -> M1(U): PHASE_U U2(OUT2) -> M1(V): PHASE_V U2(OUT3) -> M1(W): PHASE_W R2(Pin_1) -> U2(VDC): V_SENSE D1(CATHODE) -> U2(VDC): DC_BUS



Motor_Drive